

Andrew Lys

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Education

North Carolina State University Master of Financial Mathematics GPA: 4.0/4.0 Honors: Financial Mathematics Merit Fellowship Spring 2026	Raleigh, NC 12/2026
University of Chicago Bachelor of Science in Mathematics Specialization in Economics Minor in Computer Science GPA: 3.7/4.0 Honors: National Merit Scholar	Chicago, IL 06/2025

Related Coursework

NCSU PhD Courses: Bayesian Non-parametrics, Stochastic Differential Equations (SDEs), Numerical Nonlinear PDEs, Efficient Deep Learning, Statistical Theory
NCSU Master's Courses: Time Series Econometrics (Duke), Monte Carlo for Financial Math, Options and Derivatives, SQL
UChicago PhD Courses: Statistical & Computational Learning Theory, ML Theory, Geometric Methods in Computing, Combinatorics, PDEs, Functional Analysis, ML for Economists, Econometrics and ML, Causal ML, Honors Econometrics

Skills

Programming: Python, SQL, R, C++
Python Packages: NumPy, Pandas, PyTorch, JAX, PyMC, scikit-learn, XGBoost
Math: SDEs, Stochastic Control Theory, Bayesian Modeling, PDE Methods in Options Pricing, Time Series Analysis, Convex Optimization (and modern optimization/ML theory), Deep Learning, Tree Based Models, Theoretical Machine Learning.
Awards: Poker tournament winner, Putnam top 1%
Interests: Poker. Reading and doing every exercise in math books. Currently: Diffusions, Markov Processes, and Martingales vol. 1
Languages: English (native), Ukrainian (native), Russian (fluent)

Work Experience

<i>Treasury and Capital Markets – Market Transactions Quantitative Research Intern</i> Fannie Mae	06/2026-08/2026 Washington, D.C.
- Developed a novel XGBoost based model for estimating fair payups for Mortgage-Backed Security specified pools from market color and offers, for use by Fannie Mae's \$190B AUM investment portfolio to estimate hedge ratios, best execution, and theoretical pricing. - Resulted in an MAE improvement of 40% over Fannie Mae's existing internal pricing model and 42% over Bloomberg's pricing. - Used SQL and pandas to aggregate loan level data from 9+ different databases and perform data cleaning. - Designed a custom loss function to solve the problem of grouped trades and quote vs trade data.	
North Carolina State University, Financial Mathematics Department, <i>Stochastic Calculus Grader</i>	01/2026-05/2026
University of Chicago, Department of Mathematics, <i>Honors Calculus Grader</i>	09/2024-03/2024
Outlier, <i>AI Math Trainer</i>	06/2024-09/2024
University of Chicago, Department of Mathematics, <i>Research Experience for Undergrads Mentee</i>	06/2024-08/2024
University of Chicago, <i>Math Tutor</i>	09/2023-05/2024

Projects

<i>Efficient Factor-Based Conditional Diffusion Model for Contextual Portfolio Optimization</i> Financial Mathematics Department, North Carolina State University	Raleigh, NC 01/2026-05/2026
- Improved on the inference efficiency of Gao, He, et al. 2026 by implementing DDIMs for generating equities simulations. - Implemented Diffusion Transformer (DiT) and DDIM sampler from Peebles et al. and Ledoit-Wolf covariance estimator. - Resulted in 70% faster clock times, which allows our method to be scaled to intra-day portfolio optimization.	
<i>Short-Term Volatility Forecasting with Vision Transformers</i> Financial Mathematics Department, North Carolina State University	Raleigh, NC 01/2026-05/2026
Implemented image encoding of order flow data and deep learning image regression model for vol. prediction, using torch.	
<i>Introduction to Backwards Stochastic Differential Equations</i> Mathematics Department, North Carolina State University	Raleigh, NC 12/2025-12/2025
Authored an expository paper on applications of BSDEs to financial modeling of options contracts.	
<i>Multi-Alpha Trading Strategy & Portfolio Management</i> Financial Mathematics Department, North Carolina State University	Raleigh, NC 08/2025-12/2025
Constructed factor model of returns and daily trading strategy using IC. Applied model to optimize vol.-based portfolio allocation.	
<i>Weather Shocks and Airline Stock Returns: Causal Impact and Prediction</i> University of Chicago	Chicago, IL 04/2025-05/2025
Evaluated the impact of weather effects on the volatility of airline stocks. Used Random Forest, LASSO, Tensor PCA, and TFTs.	
<i>Temporal Fusion Transformers (TFTs) for Qualitative Stock Behavior</i> University of Chicago	Chicago, IL 01/2025-03/2025
Implemented transformer-based time series value trap prediction. Utilized SQL, Pandas, and PyTorch. Identified 90% of value traps.	
<i>Convolutional Neural Network (CNN) From Scratch</i> Toyota Technological Institute of Chicago	Chicago, IL 03/2025-03/2025
Wrote a CNN from scratch using only NumPy. Implemented back-propagation, striding, SGD, and more. Available on github.	
<i>Manga Sales after Anime Adaptation: Application of ML DiD</i> University of Chicago	Chicago, IL 10/2024-12/2024
Applied non-parametric DiD estimator from Hatamayar et al. 2023 to analyze the sales boost manga receive after TV adaptation.	